Infinitely many primes $3n+2$
Cambridge Tripos, 2022 IA 4-I-1E (a)

Theorem 1. *There are infinitely many primes of the form* $3n+2$ *with* $n \in \mathbb{N}$.

Proof. It suffices to prove that for every $N \in \mathbb{N}$ there exists a prime p such that $N < p$ and p is of the form $3n+2$ for some $n \in \mathbb{N}$.

We first prove the following key assertion: for all $B, M \in \mathbb{N}$, if $M \leq B$ and

$$M \equiv 2 \pmod 3,$$

then there exists a prime p such that $p \mid M$ and

$$p \equiv 2 \pmod 3.$$

We prove this by induction on B, with M universally quantified.

For $B = 0$, the assumption $M \leq B$ gives $M = 0$, which contradicts $M \equiv 2 \pmod 3$.

Now suppose the assertion holds for B, and let $M \leq B+1$ satisfy $M \equiv 2 \pmod 3$. If $M \leq B$, then the result follows immediately from the induction hypothesis. Hence assume $M \not\leq B$.

Since $M \equiv 2 \pmod 3$, we have $M \neq 1$. Therefore there exists a prime p dividing M. If $p \equiv 2 \pmod 3$, we are done. Otherwise, we show that $p \equiv 1 \pmod 3$.

Indeed, $p \not\equiv 0 \pmod 3$: if $3 \mid p$, then since $p \mid M$ we would have $3 \mid M$, contradicting $M \equiv 2 \pmod 3$. Since $p \bmod 3 < 3$, the only remaining possibility apart from 2 is

$$p \equiv 1 \pmod 3.$$

Write $M = pk$. Since $M \equiv 2 \pmod 3$ and $p \equiv 1 \pmod 3$, we get

$$k \equiv pk \equiv M \equiv 2 \pmod 3.$$

Also $k > 0$, for if $k = 0$ then $M = 0$, contradicting $M \equiv 2 \pmod 3$. Moreover, since p is prime, $p \geq 2$, and so

$$k < pk = M.$$

Thus $k < M \leq B+1$, so $k \leq B$. Also $k \mid M$, since $M = pk$. Applying the induction hypothesis to k gives a prime q such that $q \mid k$ and $q \equiv 2 \pmod 3$. Since $k \mid M$, we have $q \mid M$, which completes the induction step.

Taking $B = M$ in the key assertion, we conclude that every $M \in \mathbb{N}$ with $M \equiv 2 \pmod 3$ has a prime divisor p with $p \equiv 2 \pmod 3$.

Now let $N \in \mathbb{N}$ be given. Put

$$F = (N+1)!.$$

Then $F > 0$, and

$$A := 3F - 1 \equiv 2 \pmod 3.$$

By the key assertion, there exists a prime p such that $p \mid A$ and $p \equiv 2 \pmod 3$.

We claim that $N < p$. Suppose instead that $p \leq N$. Then $p \leq N + 1$, and since p is positive, p divides $(N + 1)! = F$. Hence $p \mid 3F$. Write

$$3F = pa, \qquad A = 3F - 1 = pb$$

for some $a, b \in \mathbb{N}$. Then

$$pa = pb + 1. \tag{1}$$

We first show that $b < a$. Otherwise $a \leq b$, so $b = a + d$ for some $d \in \mathbb{N}$. Substituting into (1) gives

$$pa = p(a + d) + 1 = pa + pd + 1,$$

which is impossible since $pd + 1 > 0$. Thus $b < a$.

Write $a = b + d$ with $d > 0$, and write $d = e + 1$ for some $e \in \mathbb{N}$. Substituting into (1) and expanding gives

$$pa = p(b + e + 1) = pb + pe + p.$$

Together with $pa = pb + 1$, this yields $1 = pe + p \geq p \geq 2$, a contradiction. Therefore the assumption $p \leq N$ is false, and so $N < p$.

Finally, since $p \equiv 2 \pmod 3$, division by 3 gives $p = 3n + 2$ where n is the quotient of p upon division by 3. Thus for every $N \in \mathbb{N}$ there is a prime $p > N$ of the form $3n + 2$ with $n \in \mathbb{N}$. Hence there are infinitely many such primes. $\square$